Final Project Proposal

Due: Tuesday, 11/23 at 11:59 p.m. EST

[GitHub]

Group Members

Alex Ding (ading13), Dylan Hu (dhu24), Geoffrey Hazard (ghazard)

Theme

We plan on rendering a number of surrealist scenes with animated cloth textured with unusual shaders in the real-time pipeline. The application allows users to switch between multiple scenes and/or the shaders used for the textiles.

Technical Features

Textiles as spring mass systems
Textile collisions with itself/other objects
Subdivision for rendering a smoother version of the cloth
Metal/glass/toon shader + texturing

Implementation Details

Cloth simulation will be implemented using a mass and spring system as described in the Matt's Webcorner article. The article provides the net force equation for each mass in the system, and it also offers various methods to integrate the equation of motion, as well as models for cloth-cloth collision and cloth-object collisions. In terms of priority, we will first want to have a functional and rendered mass-spring cloth without collision logic; we can set the initial conditions such that the cloth is deformed and restores itself to equilibrium. Then, we will incorporate cloth-object collisions to have a scene where the cloth is draped over an object, such as a sphere. All further additions to the model, such as cloth-cloth collisions and damping, would represent refinements to our cloth model, and they would be implemented with the remaining time we have.

Simultaneously, we will work on texture mapping the cloth and creating the framework to render our cloth mesh using OpenGL. This framework will include adding shaders to the cloth, as well

as algorithmic subdivision of the cloth grid such that we can simulate the cloth with a sparser grid yet still produce a smooth result.

Practically speaking, in order to simulate the cloth, we will be managing an array consisting of each mass point, with corresponding mass, acceleration and velocity.

The shaders will be implemented directly using OpenGL's shading language. Any additional time we have would be spent on creating interesting scenes.

Resources

See Spring Mass system section of these slides for theoretical / math background on spring mass systems for textiles http://www.cs.cmu.edu/~barbic/jellocube_bw.pdf
Matt's Webcorner article: https://graphics.stanford.edu/~mdfisher/cloth.html

Energy based Spring mass systems (Harder to implement given the time)
https://github.com/alecjacobson/computer-graphics-mass-spring-systems

Subdivision https://graphics.stanford.edu/~mdfisher/subdivision.html

Glass shader:
https://github.com/McNopper/OpenGL/blob/master/Example11/shader/glass.vert.glsl
https://learnopengl.com/Advanced-OpenGL/Blending
Metal shader:
https://gamedev.stackexchange.com/a/23162
Toon shader:
http://www.lighthouse3d.com/tutorials/glsl-12-tutorial/toon-shading/

Division of Labor
Dylan will work on the cloth model to begin with, including the mass-spring system and the method of integration.

Geoffrey will initially focus on the texture mapping, but will also work with Dylan to implement elements of the cloth model, such as collisions and subdivision.

Alex will implement all features unrelated to the mass-spring simulation itself, including all shaders and the code for the miscellaneous parts of the demo. With any additional time he has, he'll help out on the other components.

Plan of Action
The core component of the project is the cloth model. We plan on implementing the cloth model with increasing complexity, focusing on getting the most simple version working first. We will also be working on different features in parallel to speed up the development process: the cloth model, texture mapping (so that we can actually render the cloth), and the various shaders.

Afterwards, we will focus on improving the model with additional features and setting up scenes for the demo.